

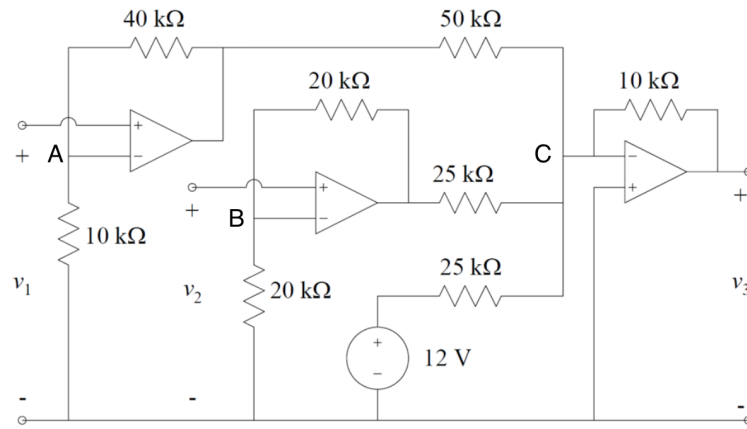
# Mechatronics HW1011

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## Problem 1

1. Please find out the output  $v_3$  as a function of the input  $v_1$ ,  $v_2$  and the Op-Amp characteristics.



For the first amplifier, we assume the a virtual short circuit,  $v_{1+} = v_{1-} = v_1$ . Apply KCL at node A,

$$\frac{0 - v_1}{10} + \frac{v_{1out} - v_1}{40} = 0$$

We get  $v_{1out} = 5v_1$ .

Same method at node B.

$$\frac{0 - v_2}{20} + \frac{v_{2out} - v_2}{20} = 0$$

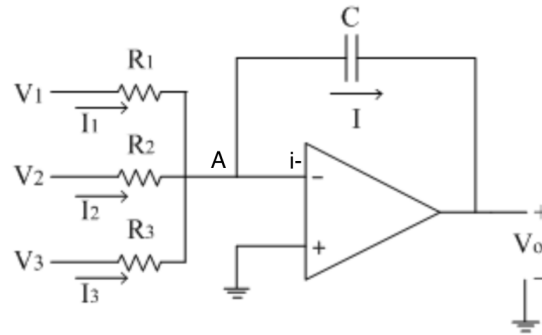
We get  $v_{2out} = 2v_2$ .

For the inverting amplifier, we assume virtual open circuit. that is  $v_- = v_+ = 0$ . Then KCL at the inverting node:

$$\begin{aligned} \frac{12 - 0}{25} + \frac{v_{1out} - 0}{50} + \frac{v_{2out} - 0}{25} &= \frac{0 - v_3}{10} \\ 25 + 5v_1 + 4v_2 &= -5v_3 \\ v_3 &= -\frac{24}{5} - v_1 - \frac{4}{5}v_2 \end{aligned}$$

## Problem 2

2. Please find out the output  $V_3$  as a function of the input  $V_1, V_2, V_3$ , and the Op-Amp characteristics.



Assume virtual open circuit at inverting node,  $i_- = 0$

KCL at node A:

$$\frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} - I = 0$$

where  $I = -C \frac{dV_0}{dt}$

$$\frac{1}{C} \left( \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) = -\frac{dV_0}{dt}$$

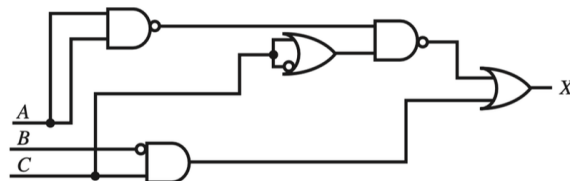
$$\int_0^t 1 dV_0 = -\frac{1}{C} \int_0^t \left( \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) dt$$

Thus,

$$V_0 = -\frac{1}{C} \int_0^t \left( \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) dt$$

## Problem 3

3. For the following circuit, please
- Write down the output  $X$  as a function of  $A, B$ , and  $C$ .
  - Simplify the expression to the simplest form.
  - Fill in the truth table.
  - Use Karnaugh map to simplify  $X$  and compare the result in b. Are they the same?



| AB |   | 00 | 01 | 11 | 10 |
|----|---|----|----|----|----|
| C  | 0 | 0  | 0  | 1  | 1  |
|    | 1 | 1  | 0  | 1  | 1  |

| A | B | C | X |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

b.

$$X = A + \overline{B} \cdot C$$

d. use the Karnaugh map to simplify  $X$ :

$$\begin{aligned} ABC\bar{C} + A\bar{B}\bar{C} + ABC + A\bar{B}C + \bar{A}\bar{B}C &= AB + A\bar{B} + BC \\ &= A + \bar{B}C \end{aligned}$$

which is same as the result compared with b

## Problem 4

4. The requirement of a logic circuit is shown in the following truth table.

- a. Please simplify the logic expression using Boolean laws.
- b. Please simplify the logic expression using Karnaugh map.
- c. Please design the simplest logic circuit to realize the functions.

| A | B | C | X |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

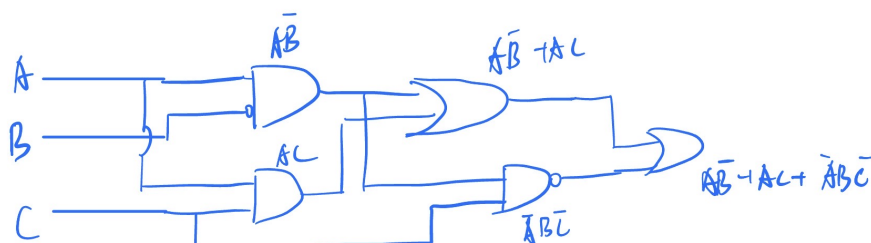
(a) Four case of  $X = 1$ ,

$$X = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}C + ABC$$

(b)

| AB |   | 00 | 01 | 11 | 10 |
|----|---|----|----|----|----|
| C  | 0 | 0  | 1  | 0  | 1  |
|    | 1 | 0  | 0  | 1  | 1  |

(c)



## Problem 5

5. Please figure out the simplest expression for the following Karnaugh map. **X** is the “don’t care term”. Hint: we did not get enough time to cover the “don’t care term”. It is a term that you can circle them if necessary. Otherwise, it is OK to leave it un-circled.

|    |    |    |    |    |    |
|----|----|----|----|----|----|
| AB |    | 00 | 01 | 11 | 10 |
| CD | 00 | 0  | 1  | 1  | 1  |
|    | 01 | 1  | 0  | 1  | 0  |
|    | 11 | 0  | x  | x  | 1  |
|    | 10 | 1  | 0  | 1  | 0  |

|    |   |    |    |    |    |
|----|---|----|----|----|----|
| AB |   | 00 | 01 | 11 | 10 |
| C  | 0 | 0  | 0  | 1  | 0  |
|    | 1 | 0  | x  | 1  | 1  |

First expression:

$$\begin{aligned}
 AB + \overline{BCD} + \overline{ACD} + ACD + \overline{ABC}D + \overline{AB}CD &= AB + \overline{AB}D + \overline{BCD} + \overline{ACD} + ACD \\
 &= A + AB + \overline{AB}D + \overline{BCD} \\
 &= A + \overline{AB}D + \overline{BCD}
 \end{aligned}$$

Second expression:

$$ABC + AB\overline{C} + A\overline{B}C = AB + AC$$